Chapter 6: Circles

Chapter 6 · Original slide 1 · Circles

Pure Y1 Ch 6
Circles

Chapter 6 · Original slide 2 · Circles

Starter (Pre-requisites)

1) Write each of the following in the form $(x+p)^2+q$:

b) $x^2+10x+28$ c) x^2-12x

c) x^2-6x+1 d) x^2+7x

5) Find the equation of the line passing through each of the following pairs of points:

f) $A(0,-6)$ and $B(4,3)$

g) $P(7,-5)$ and $Q(-9,3)$

h) $R(-4,-2)$ and $T(5,10)$

10) Use the discriminant to determine whether the following have two real solutions, one real solution or no real solutions.

k) $x^2-7x+14=0$

l) $x^2+11x+8=0$

m) $4x^2+12x+9=0$

15) Find the equation of the line that passes through the point $(3,-4)$ and is perpendicular to the line with equation $6x-5y-1=0$

Chapter 6: Circles

Chapter 6 · Original slide 3 · Chapter Overview

From this chapter onwards, the majority of the theory you will learn is new since GCSE.

1: Equation of a circle

2: Intersections of lines + circles

The diameter of a circle is AB where A and B have the coordinates $(2, 5)$ and $(8, 13)$. Determine the equation of the circle.

Show that the line $y = x - 7$ does not meet the circle $(x + 2)^2 + y^2 = 33$

NEW! since GCSE

You should already know the equation $x^2 + y^2 = r^2$ for a circle centred at the origin, but not a circle centred at a specified point.

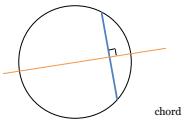
3: Chords, tangents and perpendicular bisectors.

Find the equation of the circle that passes through the points $A(-8, 1), B(4, 5), C(-4, 9)$.

4: Circumscribing Triangles

A circle C has the equation $(x - 5)^2 + (y + 3)^2 = 10$. Find the equations of the two possible tangents whose gradient is -3.

Chapter 6 · Original slide 4 · Midpoints and Perpendicular Bisectors



Later in the chapter you will need to find the perpendicular bisector of a chord of a circle.



What two properties does a perpendicular bisector of two points **A** and **B** have?

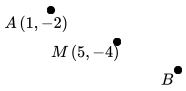
perpendicular bisector

$A(2, 5)$

Just find the mean of the x values and the mean of the y values.

Chapter 6: Circles

Chapter 6 · Original slide 5 · Test Your Understanding



Find the perpendicular bisector of the line AB where A and B have the coordinates:

a) $A\,(4, 7), B\,(10, 17)$

b) $A\,(x_1, y_1), B\,(x_2, y_2)$

A line segment AB is the diameter of a circle with centre $(5, -4)$. If A has coordinates $(1, -2)$, what are the coordinates of B?

a b

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$m_{AB} = \frac{y_2-y_1}{x_2-x_1} \qquad m_{\perp} = \frac{x_1-x_2}{y_2-y_1}$

$y - \frac{y_1+y_2}{2} = \frac{x_1-x_2}{y_2-y_1}\left(x - \frac{x_1+x_2}{2}\right)$

We could use the formula $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$, but it is easy to see from the diagram that whatever movement there is from A to M, we can continue to get to B.

Note: Do not try to memorise this!

Chapter 6 · Original slide 6 · Exercise 6A - Y1 Page 115

6 The line segment JK is a diameter of a circle, where J and K are $(\frac{3}{4}, \frac{4}{3})$ and $(-\frac{1}{2}, 2)$ respectively. The centre of the circle lies on the line segment with equation $y = 8x + b$. Find the value of b.

7 The line segment AB is a diameter of a circle, where A and B are $(0, -2)$ and $(6, -5)$ respectively. Show that the centre of the circle lies on the line $x - 2y - 10 = 0$.

Chapter 6 · Original slide 7 · Exercise 6B - Y1 Page 116

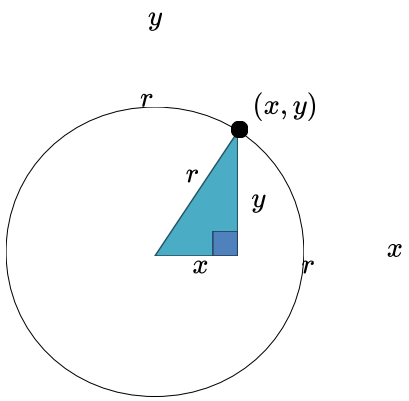
2 The line FG is a diameter of the circle centre C, where F and G are $(-2, 5)$ and $(2, 9)$ respectively. The line l passes through C and is perpendicular to FG. Find the equation of l. (7 marks)

3 The line JK is a diameter of the circle centre P, where J and K are $(0, -3)$ and $(4, -5)$ respectively. The line l passes through P and is perpendicular to JK. Find the equation of l. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

Chapter 6 · Original slide 8 · Exercise 6B - Y1 Page 117

5 Point X has coordinates $(7, -2)$ and point Y has coordinates $(4, q)$. The perpendicular bisector of XY has equation $y = 4x + b$. Find the value of q and the value of b.

Chapter 6 · Original slide 9 · Equation of a circle



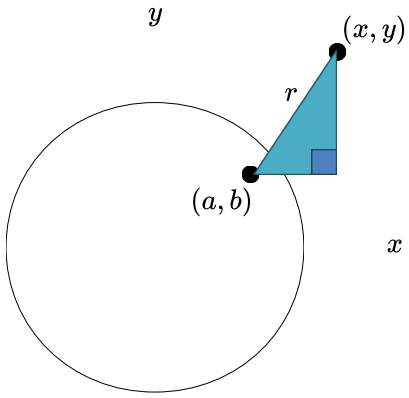
Recall that a line can be a set of points (x, y) that satisfy some equation.

Suppose we have a point (x, y) on a circle centred at the origin, with radius r. What equation must (x, y) satisfy?

(Hint: draw a right-angled triangle inside your circle, with one vertex at the origin and another at the circumference)

$x^2 + y^2 = r^2$

Chapter 6 · Original slide 10 · Equation of a circle



Now suppose we shift the circle so it's now centred at (a, b). What's the equation now?

(Hint: What would the sides of this right-angled triangle be now?)

! The equation of a circle with centre (a, b) and radius r is:

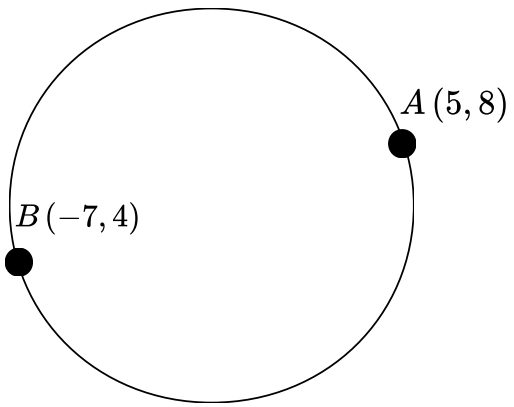
$(x - a)^2 + (y - b)^2 = r^2$

Chapter 6: Circles

Chapter 6 · Original slide 11 · Quickfire Questions

Centre	Radius
Equation	$(0,0)$
$(x+3)^2+(y-5)^2=1$	$(x+5)^2+(y-2)^2=49$
$(x+6)^2+y^2=16$	$(x-1)^2+(y+1)^2=3$
$(x+2)^2+(y-3)^2=8$	

Chapter 6 · Original slide 12 · Finding the equation using points



A line segment AB is the diameter of a circle, where A and B have coordinates $(5,8)$ and $(-7,4)$ respectively. Determine the equation of the circle.

Hint: What two things do we need to use the circle formula?

Chapter 6 · Original slide 13 · Test Your Understanding

The points A and B have coordinates $(5, -1)$ and $(13, 11)$ respectively.
(a) Find the coordinates of the mid-point of AB. (2)
Given that AB is a diameter of the circle C,
(b) find an equation for C. (4)

Chapter 6 · Original slide 14 · Completing the square

When the equation of a circle is in the form $(x-a)^2 + (y-b)^2 = r^2$, we can instantly read off the centre (a, b) and the radius r.

But what if the equation wasn't in this form?

Hint: Have we seen a method in a previous chapter that allows us to turn a x^2 term and a x term into a single expression involving x?

Complete the square!

Find the centre and radius of the circle with equation $x^2 + y^2 - 6x + 2y - 6 = 0$

Rearrange terms so that x terms are together and y terms are together.

Textbook Note: There's a truly awful method, initially presented in the textbook, that allows you to find the centre/radius without completing the square. Don't even contemplate using it.

Chapter 6 · Original slide 15 · Further Example

The circle C with centre T and radius r has equation
$x^2 + y^2 - 20x - 16y + 139 = 0$

(a) Find the coordinates of the centre of C. (3)
(b) Show that $r = 5$. (2)

Chapter 6 · Original slide 16 · Example

Given the generic equation of a circle $x^2 + y^2 + ax + by + c = 0$ find a formula to determine the co-ordinate of the centre and the radius of the circle.

$\left(x + \frac{a}{2}\right)^2 + \left(y + \frac{b}{2}\right)^2 = -c + \frac{a^2}{4} + \frac{b^2}{4}$

$r = \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 - c$

$c\left(-\frac{a}{2}, -\frac{b}{2}\right)$

Chapter 6 · Original slide 17 · Exercise 6C - Y1 Page 120

5 The line PQ is the diameter of the circle, where P and Q are $(5, 6)$ and $(-2, 2)$ respectively. Find the equation of the circle. (5 marks)

6 The point $(1, -3)$ lies on the circle $(x - 3)^2 + (y + 4)^2 = r^2$. Find the value of r.

Chapter 6 · Original slide 18 · Exercise 6C - Y1 Page 120

8 a Show that $x^2 + y^2 - 4x - 11 = 0$ can be written in the form $(x - a)^2 + y^2 = r^2$, where a and r are numbers to be found. (2 marks)
b Hence write down the centre and radius of the circle with equation $x^2 + y^2 - 4x - 11 = 0$. (2 marks)

9 a Show that $x^2 + y^2 - 10x + 4y - 20 = 0$ can be written in the form $(x - a)^2 + (y - b)^2 = r^2$, where a, b and r are numbers to be found. (2 marks)
b Hence write down the centre and radius of the circle with equation $x^2 + y^2 - 10x + 4y - 20 = 0$. (2 marks)

Chapter 6 · Original slide 19 · Exercise 6C - Y1 Page 120

12 The point $P(7, -14)$ lies on the circle with equation $x^2 + y^2 + 6x - 14y = 17$. The point Q also lies on the circle such that PQ is a diameter. Find the coordinates of point Q.

Chapter 6 · Original slide 20 · Extension

1

[MAT 2009 1B] The point on the circle $x^2 + y^2 + 6x + 8y = 75$ which is closest to the origin, is at what distance from the origin?

2

[MAT 2007 1D]

The point on the circle

$(x - 5)^2 + (y - 4)^2 = 4$

which is closest to the circle

$(x - 1)^2 + (y - 1)^2 = 1$

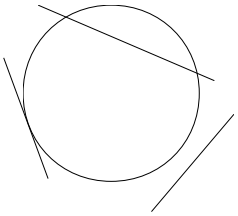
has what coordinates?

3

[MAT 2016 1I] Let a and b be positive real numbers. If

$x^2 + y^2 \leq 1$ then the largest that $ax + by$ can equal is what? Give your expression in terms of a and b.

Chapter 6 · Original slide 21 · Intersections of Lines and Circles



Show that the line $y = x + 3$ never intersects the circle with equation $x^2 + y^2 = 1$.

1 intersections (such a line is known as a tangent of the circle)

Recall that to consider the intersection of two lines, we attempt to solve them simultaneously by substitution, potentially using the discriminant to show that there are no solutions (and hence no points of intersection).

2 intersections (such a line is known as a secant of the circle)

0 intersections

Chapter 6 · Original slide 22 · Test Your Understanding

Find the points of intersection where the line $y = x + 6$ meets $x^2 + (y - 3)^2 = 29$.

Using an algebraic (and not geometric) method, determine the k such that the line $y = x + k$ touches the circle with equation $x^2 + y^2 = 1$.

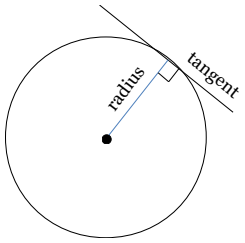
Chapter 6 · Original slide 23 · Exercise 6D - Y1 Page 122

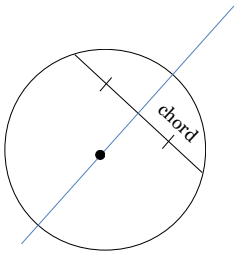
6 a Show that the line $x + y = 11$ meets the circle with equation $x^2 + (y - 3)^2 = 32$ at only one point.
(4 marks)
b Find the coordinates of the point of intersection.
(1 marks)

Chapter 6 · Original slide 24 · Exercise 6D - Y1 Page 122

11 The line with equation $y = 4x - 1$ does not intersect the circle with equation $x^2 + 2x + y^2 = k$.
Find the range of possible values of k.

Chapter 6 · Original slide 25 · Tangents, Chords, Perpendicular Bisectors





There are two circle theorems that are of particular relevance to problems in this chapter, the latter you might be less familiar with:

perpendicular bisector

The perpendicular bisector of any chord passes through the centre of the circle.

Why this will help:

The first thing we did in this chapter is find the equation of the perpendicular bisector. If we had two chords, and hence found two bisectors, we could find the point of intersection, which would be the centre of the circle.

The tangent is perpendicular to the radius (at the point of intersection).

Why this will help:

If we knew the centre of the circle and the point of intersection, we can easily find the gradient of the radius, and thus the gradient and hence equation of the tangent.

Chapter 6 · Original slide 26 · Examples



A circle C has equation
$(x-4)^2+(y+4)^2=10$

The line l is a tangent to the circle and has gradient -3. Find two possible equations for l, giving your answers in the form $y=mx+c$.

The circle C has equation
$(x-3)^2+(y-7)^2=100$.

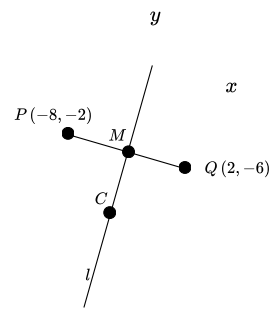
a) Verify the point $P\,(11,1)$ lies on C.

b) Find an equation of the tangent to C at the point P, giving your answer in the form $ax+by+c=0$

Tip: Use ‘subscripting’ of variables to make clear to the examiner (and yourself!) what you’re calculating.

This time we have the gradient, but don’t have the points where the tangent(s) intersect the radius.

Chapter 6 · Original slide 27 · Determining the Circle Centre



Exam Tip: If you're not asked for the equation in a particular form, just leave it as it is.

The points P and Q lie on a circle with centre C, as shown in the diagram. The point P has coordinates $(-8, -2)$ and the point Q has coordinates $(2, -6)$. M is the midpoint of the line segment PQ.

The line l passes through the points M and C.

a) Find an equation for l.

b) Given that the y-coordinate of C is -9:
i) show that the x-coordinate of C is -5.
ii) find an equation of the circle.

a b c

We know the line going through the midpoint of a chord and a centre of the circle is the perpendicular bisector of the chord.

Chapter 6 · Original slide 28 · Test Your Understanding

A circle has centre $C\,(3, 5)$, and goes through the point $P(6, 9)$. Find the equation of the tangent of the circle at the point P, giving your equation in the form $ax + by + c = 0$ where a, b, c are integers..

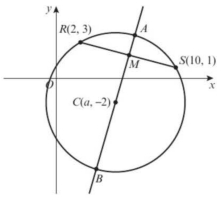
A circle passes through the points $A(0, 0)$ and $B\,(4, 2)$. The centre of the circle has x value -1. Determine the equation of the circle.

Chapter 6 · Original slide 29 · Exercise 6E - Y1 Page 126-128

6 The tangent to the circle $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.
a Find the coordinates of S and T. (5 marks)
b Hence, find the area of $\triangle OST$, where O is the origin. (3 marks)

Chapter 6 · Original slide 30 · Exercise 6E - Y1 Page 126-128

The points R and S lie on a circle with centre $C(a, -2)$, as shown in the diagram.
The point R has coordinates (2, 3) and the point S has coordinates (10, 1).
M is the midpoint of the line segment RS.
The line l passes through M and C.
a Find an equation for l. **(4 marks)**
b Find the value of a. **(2 marks)**
c Find the equation of the circle. **(3 marks)**



d Find the points of intersection, A and B, of the line l and the circle. (5 marks)

10 The points R and S lie on a circle with centre $C(a,-2)$, as shown in the diagram.
The point R has coordinates $(2,3)$ and the point S has coordinates $(10,1)$.
M is the midpoint of the line segment RS.
The line l passes through M and C.
a Find an equation for l. (4 marks)
b Find the value of a. (2 marks)
c Find the equation of the circle. (3 marks)

Chapter 6: Circles

Chapter 6 · Original slide 31 · Exercise 6E - Y1 Page 126-128

Extension:

1

[MAT 2012 1A] Which of the following lines is a tangent to the circle with equation

$x^2 + y^2 = 4$?

A) $x + y = 2$

B) $y = x - 2\sqrt{2}$

C) $x = \sqrt{2}$

D) $y = \sqrt{2} - x$

(This is not a tangent/chord question but is worthwhile regardless!)

2

[AEA 2006 Q4] The line with equation $y = mx$ is a tangent to the circle C_1 with equation $(x + 4)^2 + (y - 7)^2 = 13$

(a) Show that m satisfies the equation $3m^2 + 56m + 36 = 0$

The tangents from the origin O to C_1 touch C_1 at the points A and B.

(b) Find the coordinates of the points A and B.

Another circle C_2 has equation $x^2 + y^2 = 13$. The tangents from the point $(4, -7)$ to C_2 touch it at the points P and Q.

(c) Find the coordinates of either the point P or the point Q.

Mark scheme on next slide.

3

[STEP 2005 Q6]

(i) The point A has coordinates $(5, 16)$ and the point B has coordinates $(4, -4)$. The variable P has coordinates (x, y) and moves on a path such that $AP = 2BP$. Show that the Cartesian equation of the path of P is $(x + 7)^2 + y^2 = 100$.

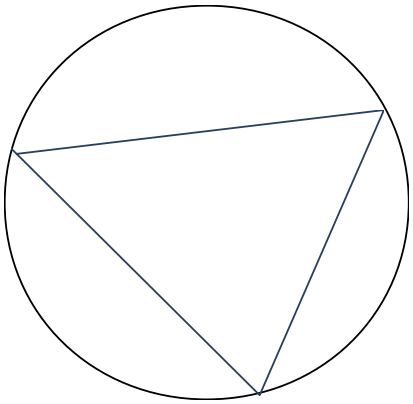
(ii) The point C has coordinates $(a, 0)$ and the point D has coordinates $(b, 0)$. The variable point Q moves on a path such that $QC = k \times QD$, where $k > 1$.
Given that the path of Q is the same as the path of P, show that $\frac{a+7}{b+7} = \frac{a^2+51}{b^2+51}$
Show further that $(a + 7)(b + 7) = 100$, in the case $a \neq b$.

Mark scheme on next slide.

Chapter 6 · Original slide 32 · Mark Scheme for Extension Question 2

Chapter 6 · Original slide 33 · Mark Scheme for Extension Question 3

Chapter 6 · Original slide 34 · Triangles in Circles



We'd say:

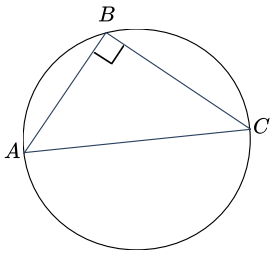
• The triangle inscribes the circle.
(A shape inscribes another if it is inside and its boundaries touch but do not intersect the outer shape)

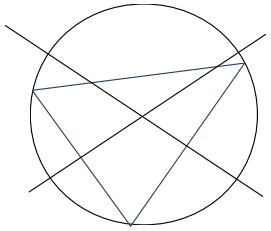
• The circle circumscribes the triangle.

• If the circumscribing shape is a circle, it is known as the circumcircle of the triangle.

• The centre of a circumcircle is known as the circumcentre.

Chapter 6 · Original slide 35 · Triangles in Circles





If $\angle ABC = 90^\circ$ then:

• **AC** is the diameter of the circumcircle of triangle **ABC**.

Similarly if AC is the diameter of a circle:

• $\angle$**ABC**$=$**90**$^\circ$ therefore **AB** is perpendicular to **BC**.

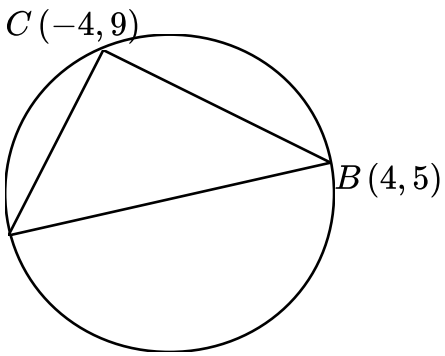
• $\mathbf{AB^2}+\mathbf{BC^2}=\mathbf{AC^2}$

Given three points/a triangle we can find the centre of the circumcircle by:

• Finding the equation of the perpendicular bisectors of two different sides.

• Find the point of intersection of the two bisectors.

Chapter 6 · Original slide 36 · Example



[Textbook] The points $A\,(-8,1), B\,(4,5), C\,(-4,9)$ lie on a circle.

a) Show that AB is a diameter of the circle.

$A\,(-8,1)$

Method 1:

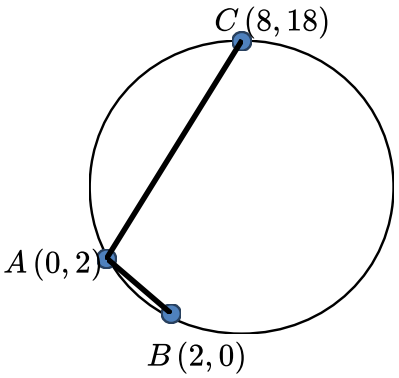
Show that $AC^2 + BC^2 = AB^2$

Method 2: (not in textbook!)

Show that AC is perpendicular to BC.

b) Hence find the equation of the circle.

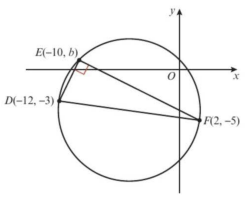
Chapter 6 · Original slide 37 · Example



The points $A\,(0,2), B\,(2,0), C\,(8,18)$ lie on the circumference of a circle. Determine the equation of the circle.

Chapter 6 · Original slide 38 · Exercise 6F - Y1 Page 131-132

8 The points $D(-12,-3), E(-10,b)$ and $F(2,-5)$ lie on the circle C as shown in the diagram.
Given that $\angle DEF = 90°$ and $b > 0$
a show that $b = 1$ (5 marks)
b find an equation for C. (4 marks)



Chapter 6 · Original slide 39 · Exercise 6F - Y1 Page 131-132

9 A circle has equation
$x^2 + 2x + y^2 - 24y - 24 = 0$.
a Find the centre and radius of the circle.
b The points $A(-13, 17)$ and $B(11, 7)$ both lie on the circumference of the circle. Show that AB is a diameter of the circle.
c The point C lies on the negative x-axis and the angle $ACB = 90°$. Find the coordinates of C.

Chapter 6 · Original slide 40 · Exercise 6F - Y1 Page 131-132

Extension:

1

[STEP 2009 Q8 Edited] If equation of the circle C is $(x - 2t)^2 + (y - t)^2 = t^2$, where t is a positive number, it can be shown that C touches the line $y = 0$ as well as the line $3y = 4x$.

Find the equation of the incircle of the triangle formed by the lines $y = 0, 3y = 4x$ and $4y + 3x = 15$.

Note: The incircle of a triangle is the circle, lying totally inside the triangle, that touches all three sides.